

Week-2

Rayanki Himavarshini(2520030141)

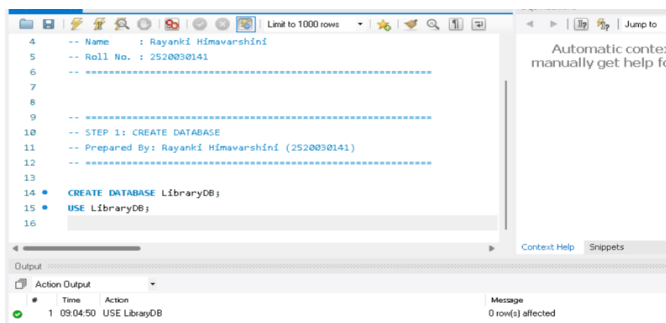
Week-2

The Librarian's Dashboard Scenario The library is now digital! The head librarian needs a report to see which members are active and which books are in the collection. You also need to handle a "Book Donation" process where a new book is added and assigned to a category at the exact same time—if one part fails, the whole process must stop to keep the catalog accurate.

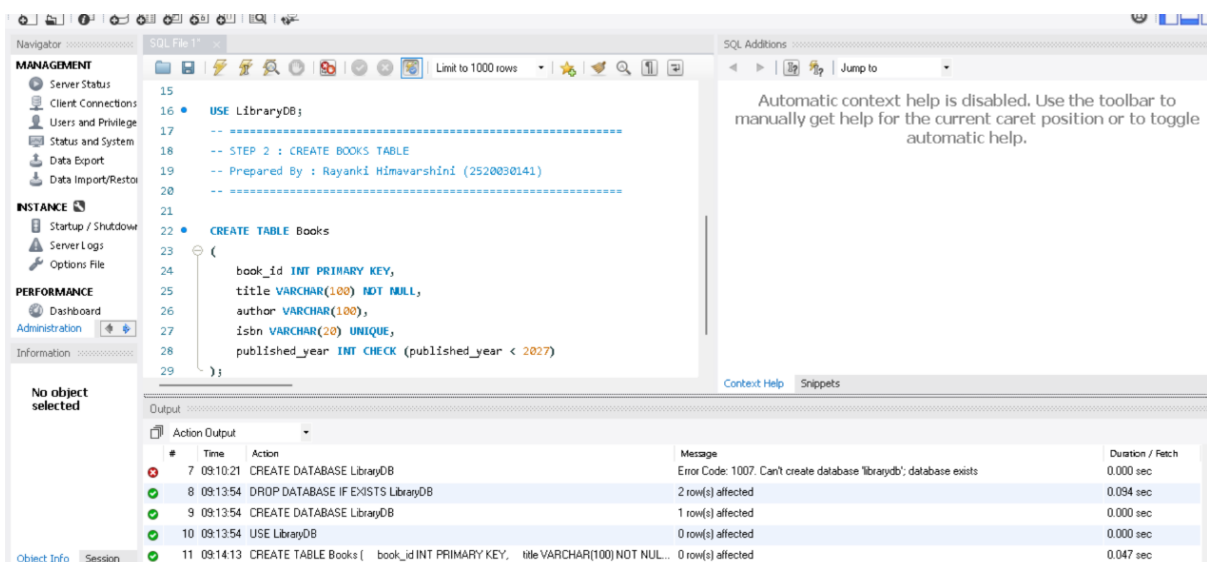
Task to be Done

1. The Catalog Search (JOINS): Create a third table called Loans that connects member_id to book_id. Write a query to show the Member Name and the Book Title they have currently borrowed.
2. Collection Stats (Aggregations): Write a query to find the total number of books published in each year. Hint: Use COUNT(book_id) and GROUP BY published_year.
3. The Donation Transaction (ACID): Use a Transaction (BEGIN...COMMIT) to add a new book to the Books table and simultaneously log the entry in a Donation_History table.
4. Search Speed (Indexing): Create an Index on the isbn column. Explain why searching by ISBN is now faster for the librarian than searching by Title. Skill Set Gained • Relational Logic: Connecting independent tables (Books + Members) through a Junction table (Loans). • Analytical SQL: Using GROUP BY to generate business insights. • Atomic Operations: Using Transactions to ensure multiple database changes succeed or fail as a single unit.

Step 1 — Create Database



Step 2 — Create Books Table (DDL + Constraints)



Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

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SQL File 1 x

Limit to 1000 rows

```

41
42 • SELECT * FROM Books;
43
44 -- STEP 4 : CREATE MEMBERS TABLE
45 -- Prepared By : Rayanki Himavarshini (2520030141)
46 --
47
48 • CREATE TABLE Members
49 (
50     member_id INT PRIMARY KEY,
51     full_name VARCHAR(100),
52     email VARCHAR(100) UNIQUE
53 );
54
55

```

SQL Additions

Automatic context help is disabled. Use manually get help for the current caret position or automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message
✓ 43	09:17:35	USE LibraryDB	0 row(s) affected
✓ 44	09:17:35	CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100) NOT NU...	0 row(s) affected
✓ 45	09:17:35	INSERT INTO Books (book_id,title,author,isbn,published_year) VALUES (1,'The Great Bo...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 46	09:17:35	SELECT * FROM Books LIMIT 0,1000	3 row(s) returned
✓ 47	09:17:35	CREATE TABLE Members (member_id INT PRIMARY KEY, full_name VARCHAR(10...	0 row(s) affected

SQL File 1 x

Limit to 1000 rows

Result Grid

book_id	title	author	isbn	published_year
1	The Great Book	F. Scott Fitzgerald	9780743273565	1925
2	To Kill a Mockingbird	Harper Lee	9780061120084	1960
3	1984	George Orwell	9780451524935	1949
* NULL	NULL	NULL	NULL	NULL

Books 6 x

Apply Revert

Context Help Snippets

Output

Action Output

#	Time	Action	Message
✓ 52	09:18:10	USE LibraryDB	0 row(s) affected
✓ 53	09:18:10	CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100) NOT NU...	0 row(s) affected
✓ 54	09:18:10	INSERT INTO Books (book_id,title,author,isbn,published_year) VALUES (1,'The Great Bo...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 55	09:18:10	SELECT * FROM Books LIMIT 0,1000	3 row(s) returned
✓ 56	09:18:11	CREATE TABLE Members (member_id INT PRIMARY KEY, full_name VARCHAR(10...	0 row(s) affected

SQL File 1*

```

52  );
53  --
54  -- STEP 5 : INSERT MEMBER RECORDS
55  -- Prepared By : Rayanki Himavarshini (2520030141)
56  --
57  • INSERT INTO Members
58    (member_id, full_name, email)
59  VALUES
60    (101, 'Rayanki Himavarshini', '2520030141@klh.edu.in'),
61    (102, 'Emma Wilson', 'emma@gmail.com'),
62    (103, 'Michael Brown', 'michael@gmail.com');
63
64  • SELECT * FROM Members;
65
66  --

```

Automatic context help is disabled. Use the manually get help for the current caret position or automatic help.

Output

#	Time	Action	Message
✓ 50	09:18:10	DROP DATABASE IF EXISTS LibraryDB	2 row(s) affected
✓ 51	09:18:10	CREATE DATABASE LibraryDB	1 row(s) affected
✓ 52	09:18:10	USE LibraryDB	0 row(s) affected

SQL File 1*

```

67  -- STEP 6 : CREATE LOANS TABLE
68  -- Prepared By : Rayanki Himavarshini (2520030141)
69  --
70  • CREATE TABLE Loans
71  (
72    loan_id INT PRIMARY KEY,
73    member_id INT,
74    book_id INT,
75    loan_date DATE,
76    FOREIGN KEY(member_id) REFERENCES Members(member_id),
77    FOREIGN KEY(book_id) REFERENCES Books(book_id)
78  );
79
80  --
81  -- STEP 7 : INSERT LOAN RECORDS

```

Automatic context help is disabled. Use the manually get help for the current caret position or automatic help.

Output

#	Time	Action	Message
✓ 50	09:18:10	DROP DATABASE IF EXISTS LibraryDB	2 row(s) affected
✓ 51	09:18:10	CREATE DATABASE LibraryDB	1 row(s) affected
✓ 52	09:18:10	USE LibraryDB	0 row(s) affected
✓ 53	09:18:10	CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100) NOT NU...	0 row(s) affected
✓ 54	09:18:10	INSERT INTO Books (book_id,title,author,isbn,published_year)VALUES (1,'The Great Bo...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0

SQL File 1*

```

46
47 CREATE TABLE Members
48 (
49     member_id INT PRIMARY KEY,
50     full_name VARCHAR(100),
51     email VARCHAR(100) UNIQUE

```

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid

loan_id	member_id	book_id	loan_date
1	101	1	2025-01-05
2	102	2	2025-01-08
3	103	3	2025-01-10
4	101	2	2025-02-01
5	102	1	2025-02-05
NULL	NULL	NULL	NULL

Books 7 Members 8 Loans 9

Output

Action Output

#	Time	Action	Message	Duration / Fetch
65	09:20:09	INSERT INTO Members (member_id,full_name,email) VALUES (101,Rayanki Himavarshini...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec
66	09:20:09	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
67	09:20:09	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id IN...	0 row(s) affected	0.062 sec
68	09:20:10	INSERT INTO Loans (loan_id,member_id,book_id,loan_date)VALUES (1,101,1,2025-01-...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec
69	09:20:10	SELECT * FROM Loans LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec

135

Result Grid

book_id	title	author	isbn	published_year
1	The Great Book	F. Scott Fitzgerald	9780743273565	1925
2	To Kill a Mockingbird	Harper Lee	9780061120084	1960
3	1984	George Orwell	9780451524935	1949
NULL	NULL	NULL	NULL	NULL

Books 10

Output

Action Output

#	Time	Action	Message
66	09:20:09	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned
67	09:20:09	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id IN...	0 row(s) affected
68	09:20:10	INSERT INTO Loans (loan_id,member_id,book_id,loan_date)VALUES (1,101,1,2025-01-...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
69	09:20:10	SELECT * FROM Loans LIMIT 0, 1000	5 row(s) returned
70	09:20:55	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned

131 donor_name VARCHAR(100),

132 donation_date DATE,

133

134 FOREIGN KEY(book_id) REFERENCES Books(book_id)

135);

136

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid

Member_Name	Book_Title
Rayanki Himavarshini	The Great Book
Rayanki Himavarshini	To Kill a Mockingbird
Emma Wilson	To Kill a Mockingbird
Emma Wilson	The Great Book
Michael Brown	1984

Books 16 Members 17 Loans 18 Result 19 Result 20

Output

Action Output

#	Time	Action	Message	Duration / Fetch
97	09:21:29	INSERT INTO Loans (loan_id,member_id,book_id,loan_date)VALUES (1,101,1,2025-01-...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.015 sec
98	09:21:29	SELECT * FROM Loans LIMIT 0, 1000	5 row(s) returned	0.000 sec
99	09:21:29	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans INNER JOI...	5 row(s) returned	0.000 sec
100	09:21:29	SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP BY pu...	3 row(s) returned	0.000 sec
101	09:21:29	CREATE TABLE Donation_History (donation_id INT PRIMARY KEY, book_id INT, ...	0 row(s) affected	0.032 sec

134 FOREIGN KEY(book_id) REFERENCES Books(book_id)
135);
136

Result Grid

	published_year	Total_Books
▶	1925	1
	1949	1
	1960	1

Books 16 Members 17 Loans 18 Result 19 Result 20 x Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message
✓ 97	09:21:29	INSERT INTO Loans (loan_id,member_id,book_id,loan_date) VALUES (1,101,1,'2025-01-...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
✓ 98	09:21:29	SELECT * FROM Loans LIMIT 0, 1000	5 row(s) returned
✓ 99	09:21:29	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l INNER JOI...	5 row(s) returned
✓ 100	09:21:29	SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP BY pu...	3 row(s) returned
✓ 101	09:21:29	CREATE TABLE Donation_History (donation_id INT PRIMARY KEY, book_id INT, ...	0 row(s) affected

Result Grid

	donation_id	book_id	donor_name	donation_date
▶	1	4	Raj Kumar	2026-08-07
*	NULL	NULL	NULL	NULL

Donation_History 31 Books 32 Books 33 Members 34 Loans 35 Donati Apply Revert Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 138	09:23:59	SELECT * FROM Books WHERE isbn='9780451526342' LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.
✓ 139	09:23:59	SELECT * FROM Books LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.
✓ 140	09:23:59	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.
✓ 141	09:23:59	SELECT * FROM Loans LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.
✓ 142	09:23:59	SELECT * FROM Donation_History LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.

Result Grid

	donation_id	book_id	donor_name	donation_date
▶	1	4	Raj Kumar	2026-08-07
*	NULL	NULL	NULL	NULL

Donation_History 42 x Books 43 Books 44 Members 45 Loans 46 Apply Revert Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 166	09:25:00	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
✓ 167	09:25:00	SELECT * FROM Loans LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
✓ 168	09:25:00	SELECT * FROM Donation_History LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

151 -- ROLLBACK;
152 -----

Result Grid

book_id	title	author	isbn	published_year
4	Animal Farm	George Orwell	9780451526342	1945

Output

Action Output

#	Time	Action	Message	Duration / Fetch
166	09:25:00	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
167	09:25:00	SELECT * FROM Loans LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
168	09:25:00	SELECT * FROM Donation_History LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

SQL File 1

```
147 VALUES  
148 (1,4,'Raj Kumar',CURDATE());  
149 COMMIT;  
150 -- If any error occurs  
151 -- ROLLBACK;  
152 -----
```

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid

book_id	title	author	isbn	published_year
1	The Great Book	F. Scott Fitzgerald	9780743273565	1925
2	To Kill a Mockingbird	Harper Lee	9780061120084	1960
3	1984	George Orwell	9780451524935	1949
4	Animal Farm	George Orwell	9780451526342	1945

Output

Action Output

#	Time	Action	Message	Duration / Fetch
151	09:25:00	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
152	09:25:00	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id INT, ...	0 row(s) affected	0.031 sec
153	09:25:00	INSERT INTO Loans (loan_id,member_id,book_id,loan_date) VALUES (1,101,1,2025-01-...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec
154	09:25:00	SELECT * FROM Loans LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench

mySQL x

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

- Server Status
- Client Connections
- Users and Privilege
- Status and System
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Administration

Information

No object selected

SQL File 1

Output

Action Output

#	Time	Action	Message	Duration / Fetch
97	09:21:29	INSERT INTO Loans (loan_id,member_id,book_id,loan_date) VALUES (1,101,1,2025-01-...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.015 sec
98	09:21:29	SELECT * FROM Loans LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
99	09:21:29	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans INNER JOIN...	5 row(s) returned	0.000 sec / 0.000 sec
100	09:21:29	SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP BY pu...	3 row(s) returned	0.000 sec / 0.000 sec
101	09:21:29	CREATE TABLE Donation_History (donation_id INT PRIMARY KEY, book_id INT, ...	0 row(s) affected	0.032 sec
102	09:22:17	DROP DATABASE IF EXISTS LibraryDB	4 row(s) affected	0.078 sec
103	09:22:17	CREATE DATABASE LibraryDB	1 row(s) affected	0.016 sec
104	09:22:17	USE LibraryDB	0 row(s) affected	0.000 sec
105	09:22:17	CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100) NOT NU...	0 row(s) affected	0.062 sec
106	09:22:17	INSERT INTO Books (book_id,title,author,isbn,published_year) VALUES (1,'The Great Bo...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec
107	09:22:17	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
108	09:22:17	CREATE TABLE Members (member_id INT PRIMARY KEY, full_name VARCHAR(10...	0 row(s) affected	0.032 sec
109	09:22:17	INSERT INTO Members (member_id,full_name,email) VALUES (101,'Rajanki Himavashini'...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.015 sec
110	09:22:17	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
111	09:22:17	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id IN...	0 row(s) affected	0.047 sec
112	09:22:17	INSERT INTO Loans (loan_id,member_id,book_id,loan_date) VALUES (1,101,1,2025-01-...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.015 sec
113	09:22:17	SELECT * FROM Loans LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
114	09:22:17	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans INNER JOIN...	5 row(s) returned	0.000 sec / 0.000 sec
115	09:22:17	SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP BY pu...	3 row(s) returned	0.000 sec / 0.000 sec
116	09:22:17	CREATE TABLE Donation_History (donation_id INT PRIMARY KEY, book_id INT, ...	0 row(s) affected	0.032 sec
117	09:23:59	DROP DATABASE IF EXISTS LibraryDB	4 row(s) affected	0.094 sec

Management

- Server Status
- Client Connections
- Users and Privilege
- Status and System
- Data Export
- Data Import/Restore

STANDBY

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Administration

No object selected

SQL File 1*

Limit to 1000 rows

SQL Additions

- Jump to

Automatic context help is disabled. Use the tool manually get help for the current caret position or automatic help.

Result Grid

Filter Rows:

Edit: Export/Import:

Form Editor

member_id	full_name	email
101	Rayanki Himavarshini	2520030141@kjh.edu.in
102	Emma Wilson	emma@gmail.com
103	Michael Brown	michael@gmail.com
NULL	NULL	NULL

Donation_History 42 Books 43 Books 44 Members 45 Loans 46 Apply Revert Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration
97	09:21:29	INSERT INTO Loans (loan_id,member_id,book_id,loan_date) VALUES (1,101,1,'2025-01-...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.015
98	09:21:29	SELECT * FROM Loans LIMIT 0, 1000	5 row(s) returned	0.000
99	09:21:29	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l INNER JOIN...	5 row(s) returned	0.000
100	09:21:29	SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP BY pu...	3 row(s) returned	0.000